TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Dr. Benjamin Heyne	Email	Benjamin.heyne@iap.fraunhofer.de
Institute Name	Fraunhofer Institute for Applied Polymer Research IAP			
Address	Geiselbergstr. 69, 14476 Potsdam			

2. Technology Overview

Title of Proposed Technology	Sustainable CCMs - Multielement catalysts on PFAS-free membranes as plattform technology for the hydrogen-based industry (SustainCCM)			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		**Chemicals**	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	Hydrogen	Polymeric Materials	Materials for Energy/Environmental Industries
Concept of Proposed Technology	**Concept of Proposed Technology** • **Core concept:** Development of **PFAS-free polymer anion exchange membranes (AEM)** for Anion Exchange Membrane Water Electrolyzers (AEMWE), paired with nanoscale **PGM-reduced/precious metal free multi-element catalysts**, **ionomer integration**, and scalable processing up to stack level. • **Main functions:** ○ Membranes with **high ionic conductivity**, **chemical resistance**, and **mechanical stability**. ○ Non-PGM catalysts optimized for **HER** and **OER** with synergistic metal interactions to boost activity and reduce poisoning effects.			

(max 10 pages)

	○ Uniform catalyst coatings via layer-by-layer deposition and in situ synthesis for enhanced adhesion and minimal loading. ○ Engineered membrane morphology for enhanced ion transport and reduced gas cross-over • **Potential uses:** ○ **AEMWE stack testing** (in collaboration with GHS) to validate full-system performance, efficiency, and durability. ○ **Cost-efficient up-scaling** from batch to continuous, e.g. **roll-to-roll, processing** for industrial membrane/catalyst production. ○ Sustainable **green hydrogen production** enabling eco-friendly energy conversion with reduced material costs and environmental impact. ○ Potential additional uses of materials and processing for applications in carbon capture and utilization technologies
Rationale for Joint R&D Project	• **Urgent market need**: Global hydrogen economy growth demands **scalable, cost-effective AEMWE systems** to meet DOE targets (e.g., 500 mA/cm² at 1.8 V), but current technologies rely on **PFAS-based membranes** (environmental risks) and **PGM catalysts** (high cost, supply limits). • **Korean industry leadership**: Korea dominates **PEMWE stack manufacturing** and electrolyzer deployment, but lacks expertise in **PFAS-free AEM materials** and **non-PGM catalysts** needed to reduce CAPEX by >50% and enter alkaline markets. • **Strategic gap**: Korean partners provide **stack engineering, upscaling infrastructure, and market access**, while Fraunhofer IAP delivers PFAS-Free **polymer synthesis, adapted membrane morphology** and **nanocatalyst development** — complementary competencies essential for full-system breakthrough. • **Economic impact**: Project enables Korean industry to **diversify from PEM/PGM dependency**, capture **AEMWE market share** (projected >$10B by 2030), and lead sustainable hydrogen production using abundant materials. • **Collaboration necessity**: Only through **German materials science + Korean systems integration** can PFAS-free MEAs be rapidly scaled from lab to MW-class stacks, accelerating commercialization and regulatory compliance (REACH/PFAS bans).
Key Words	1. PFAS-free anion exchange membrane 2. Transport optimized membrane morphology 3. PGM-reduced/ noble metal-free catalysts 4. Catalyst coated membrane (CCM) 5. Fuel cell or electrolyzer

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (8)
R&D Objectives	• PFAS-free membranes in connection with multi-element, PGM-reduced/ noble metal-free catalysts • PFAS-free ion conductive polymers, Optimized membrane processing and PGM-reduced catalysts • scale-up of polymer synthesis, processing of membranes and scale-up, continuous producing of the nanoscale catalysts • cost assessment, business plan and commercialization plan			
R&D Target Specifications	**This study combines material development and process technologies up to system level in the field of hydrogen technologies.** The project goal is the development of perfluoroalkyl substance (PFAS)-free membranes for anion exchange membranes (AEM) and/ or proton exchange membranes (PEM) in connection with multi-element, PGM reduced/ noble metal-free catalysts with high performance and increased stability in electrochemical applications, and their processing up to the system level. Traditional membranes often contain PFAS, raising environmental and health concerns. We investigate alternative materials and fabrication techniques (from batch wise membrane fabrication to roll-2-roll processing) that ensure high ionic conductivity, mechanical stability, and chemical resistance without the use of PFAS. The performance of these membranes is assessed in terms of efficiency and durability in various electrochemical environments. Our findings highlight the potential of PFAS-free membranes to meet the growing demand for sustainable and eco-friendly solutions in energy conversion and storage technologies. PGM is the most used catalyst material in PEM and is even found in AEM technologies. Reducing the scarce and expensive elements is key for reducing costs and increasing market share for building up hydrogen industry. The interaction between the metals incorporated in the new catalyst particles show beneficial interactions within the electrocatalytic reactions and help boosting the catalytic reaction and reducing poisoning. Building on the development of PFAS-free membranes and catalyst particle manufacturing, this section examines the advancements in membrane development and catalyst coating techniques and upscaling of said materials. **One focus** is on optimizing the membrane structure to enhance ion transport and reduce resistance, thereby improving overall performance in AEM and PEM systems. We explore various polymer blends and composite materials that provide enhanced mechanical properties and ionic conductivity. Based on the competencies in the nanotechnologies, t**he second focus** is on the manufacturing of multi-element catalyst nanoparticles with precise composition, size and shape control for enhanced conversion rates. Followed by the **third focus** on the cost-efficient production of morphology optimized membranes to adjust membrane properties like mechanical stability permeability. Furthermore, we investigate innovative catalyst coating methods, including layer-by-layer deposition and in situ synthesis, to achieve uniform catalyst distribution on the membrane surface and enhanced adhesion in order to reduce degradation, resulting in long-term stability which will further reduce costs. These coatings aim to maximize active sites and current and ion conductivity while minimizing catalyst loading, thus improving efficiency and reducing costs. The synergistic effects of optimized membranes and advanced catalyst coatings are evaluated, showing promising results for enhanced electrochemical performance and longevity in fuel cells and electrolyzers. This integrated approach paves the way for the next generation of sustainable and high-performance electrochemical devices.			

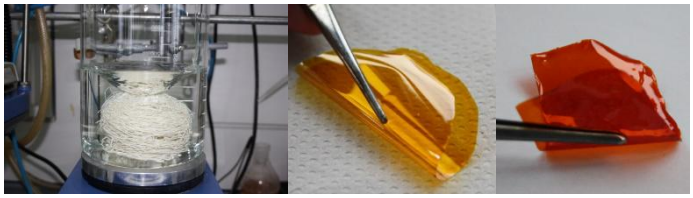
(max 10 pages)

R&D Plan	**Research Approach**
	• Integrated development from **materials synthesis** to **system-level stack testing**, focusing on PFAS-free polymer membranes, non-PGM catalysts, and scalable processing.
	• Iterative cycles of **material optimization**, **electrochemical testing**, and **upscaling** with performance validation at each stage.
	• Collaboration with industrial partner (GHS) for stack integration and real-world testing.
	Key Tasks
	• **Task 1: PFAS-free membrane development**
	  ○ Synthesis of polymer backbones and anion exchange ionomers with high ionic conductivity.
	  ○ Optimization of mechanical/chemical stability via crosslinking and composite fillers.
	  ○ Membrane morphology design, to improve permeability and reduce gas crosspver
	  ○ Scale-up from lab casting to roll-to-roll processing.
	• **Task 2: Non-PGM multi-element catalyst synthesis**
	  ○ Controlled colloidal nanoparticle production (size, shape, composition) using nanotechnology expertise.
	  ○ Formulation of PGM-reduced/noble metal-free alloys with synergistic metal interactions for HER/OER.
	  ○ Ionomer integration for improved catalyst-membrane interface.
	• **Task 3: Catalyst coating and MEA fabrication**
	  ○ Development of layer-by-layer deposition and in situ synthesis methods.
	  ○ Optimization of catalyst loading, uniformity, and adhesion on PFAS-free membranes.
	  ○ Assembly of membrane-electrode assemblies (MEAs) with minimal resistance.
	• **Task 4: Electrochemical testing and stack validation**
	  ○ Single-cell testing (efficiency, durability, current density).
	  ○ Full AEMWE stack testing with GHS partner.
	  ○ Long-term stability assessment under operational conditions.
	Methodologies
	• **Materials synthesis**: Polymerization, sol-gel processes, nanoparticle synthesis via wet chemistry.
	• **Processing**: Solution casting, slot-die coating, roll-to-roll pilot line.
	• **Characterization**: Ionic conductivity (EIS), mechanical testing (tensile, DMA), chemical stability (soaking tests), microscopy (SEM/TEM).
	• **Electrochemical testing**: Half-cell RDE, full-cell polarization curves, accelerated stress tests (AST), stack performance mapping.
	Expected Milestones and Timeline (24 months)
	Months 1-6: Material foundation
	• PFAS-free ionomer formulations with >50 mS/cm conductivity at 60°C.
	• First non-PGM catalyst nanoparticles synthesized and characterized.
	• Initial MEA prototypes fabricated.
	Months 7-12: Performance optimization
	• Membranes demonstrating >1000 h stability in 1M KOH.

	• Catalyst coatings achieving >2 A/cm² at 1.8 V cell voltage. • Roll-to-roll processing parameters established. **Months 13-18: System integration** • Optimized MEAs with >70% efficiency at 1 A/cm². • First AEMWE stack assembly and initial testing with GHS. • Techno-economic analysis completed. **Months 19-24: Validation and upscaling** • Stack durability >5000 h at 0.5 A/cm² with <10 μV/h degradation. • Final roll-to-roll membrane production validated (>10 m²/h). • Performance report and scale-up roadmap for industrial implementation.

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Development of novel PFAS-free materials for electrolysers or fuel cells; Modfication and optimization on molecular level; Membrane production, its optimization and modification up to catalyst application with reduced noble metal content Examples of PFAS-free material and membranes  ○ Insights on surface chemistry modification of developed materials for enhanced stability and catalytic properties using specific characterization methods ○ Processing of developed materials in order to manufacture CCM/MEA components with the incorporated the non-noble metal catalysts for application on system level and characterization ○ Development of roll-2-roll manufacturing processes and parameters to scale up the membrane fabrication
Expected R&R of Korean SME Partner	Commercialization of our developed product → Industrialization of PFAS-free membranes within the electrolyser and fuel cell community → production of enhanced colloidal catalyst material, i.e. application-specific catalyst particles and support material (carbons and oxides) → manufacturing technology for production upscaling for multi-element, PGM reduced catalysts for electrocatalysis (from lab to fab) → manufacturing technology for producing industrial scale membrane, catalysts and CCMs

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	○ Electrolyzers and fuel cells in field tests ⬭ data on performance for further (machine learning/KI based) optimization ⬭ commercially available enhanced catalyst material

6. References

(1) S. Janietz a, T. Brening a, Th. Schiestel b, T. Götz; Block copoly(phenylquinoxaline)s as potential ionomers for proton exchange membranes; doi.org/10.1016/j.polymer.2023.125794

(2) Patent grant: Process for the preparation of block copolyquinoxalines with sulphonic or phosphonic acid groups and their use as proton-conducting membranes; DE 10 2017 208 805.2

(3) Patent grant: Block copoly(phenylquinoxalines) with quaternary groups and their use as alkaline conductive membranes;

DE 10 2019 203 597.3

(4) A. Karaca, A. Glüsen, K. Wippermann, S. Mauger, A. C. Yang-Neyerlin, S. Woderich, C. Gimmler, M. Müller, G. Bender, H. Weller, M. Carmo, and D. Stolten; Oxygen Reduction at PtNi Alloys in Direct Methanol Fuel Cells – Electrode Development and Characterization; Energies **2023**, 16,1115; DOI: 10.3390/en16031115

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization